Aarav Shah

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ABOUT ME

I engineer intelligent systems that bridge theoretical rigor with real-world impact—from quantum-inspired supply chain optimization to accessibility technology for users with disabilities. My work sits at the intersection of applied mathematics, machine learning, and operations research, driven by the conviction that the most elegant algorithms are those that solve problems people actually face.

ACADEMIC PROFILE

University of California, Riverside
Key Interests: Operations Research, Machine Learning Systems, Applied Mathematics
Research Focus: Quantum decision theory in supply chains; Neural representations for scientific data

LEADERSHIP & SIGNIFICANT ACTIVITIES

Quantum-Classical Supply Chain Research | Lead Researcher

2024–Present • 40 hrs/week • 2,080 total hours
Aligns with MIT's: Operations Research Center (Prof. Bertsimas's prescriptive analytics)

When COVID-19 disrupted global supply chains, I watched companies lose billions to demand volatility they couldn't predict. This sparked my investigation into whether quantum decision theory could capture customer behavior that classical models miss. I developed a unified framework integrating ARIMA forecasting with quantum superposition modeling of customer intent, embedding forecast uncertainty directly into reinforcement learning reward structures. The system achieved 12–18% profitability improvements in simulated omnichannel retail environments while reducing stock-outs by 23%. This work extends classical operations research into quantum regimes—precisely the frontier Prof. Bertsimas explores at MIT's ORC.

Key Outcomes:
- 12–18% profitability improvement over classical baselines
- 23% reduction in stock-out events under demand volatility
- IEEE publication documenting novel quantum-RL integration

Hyperspectral Data Research | Research Assistant, UC Riverside

2024–Present • 20 hrs/week • 520 total hours
Aligns with MIT's: CSAIL computational imaging research

I joined this project to answer a fundamental question: when do neural implicit representations outperform classical tensor decomposition for high-dimensional scientific data? I implemented SVD, CP, and Tucker decompositions alongside custom coordinate-based neural networks, designing controlled experiments across compression ratios, noise levels, and model capacities.

My analysis revealed that neural models achieve superior reconstruction at >10x compression while classical methods maintain stability advantages under noise. This comparative framework now guides method selection for remote sensing applications.

Key Outcomes:
• Benchmarked 5 decomposition methods across 12 experimental conditions
• Identified 10x compression threshold where neural models dominate
• Built reproducible evaluation framework for scientific imaging

Curriculum Manager | Daricha Organization

April 2024–Present • 25 hrs/week • 1,950 total hours
Aligns with MIT's: "Mens et Manus" – technical skills for humanitarian impact

After the Taliban banned girls from schools in Afghanistan, I couldn't accept that millions would lose access to education. I joined Daricha to design curricula deliverable through covert channels—adapting content for low-bandwidth environments and intermittent access. I restructured the mathematics and science programs to function in 15-minute modular segments, enabling learning during brief safe windows. The curriculum now reaches 400+ girls across 12 Afghan provinces, with 89% completion rates despite extraordinary constraints.

Key Outcomes:
• Designed modular curriculum reaching 400+ girls in 12 provinces
• Achieved 89% course completion rate in conflict zone
• Adapted 200+ hours of content for low-bandwidth delivery

F1 Race Insights | Founder & Lead Developer

2024–Present • 15 hrs/week • 200 total hours
Aligns with MIT's: CSAIL end-to-end ML systems approach

I built this to prove that rigorous probabilistic modeling could match professional F1 team predictions. I engineered 68 features from 1,000+ historical races, implemented 8 ML models including a custom neural architecture (NBT-TLF), and deployed 10,000 Monte Carlo simulations for championship projections. The system achieves 98.7% AUC-ROC—14x better than random chance—with SHAP explainability and counterfactual analysis. Full-stack deployment includes Python/FastAPI backend, Next.js frontend, Docker containers, and AWS infrastructure with drift monitoring.

Key Outcomes:
• 98.7% AUC-ROC accuracy across 3 seasons of predictions
• 8 ML models benchmarked with automated A/B testing
• Production deployment with MLOps monitoring pipeline

AI Implementation Engineer | SAP

May 2025–December 2025 • 20 hrs/week • 500 total hours
Aligns with MIT's: MIT-IBM Watson AI Lab enterprise deployment

I translated experimental AI models into production-ready services within SAP's enterprise infrastructure. I built monitoring, logging, and rollback mechanisms for workforce-facing applications, then iterated based on real-world error analysis rather than offline metrics. My work reduced the gap between prototype and production performance by 34%, establishing deployment pipelines for continuous AI iteration across SAP's enterprise software suite.

Key Outcomes:
• Deployed 6 ML models to production with <50ms latency
• Reduced prototype-to-production performance gap by 34%
• Built validation framework adopted across 3 product teams

Inventory Quantum | Founder & Developer

January 2025–Present • 20 hrs/week • 300 total hours
Aligns with MIT's: Sloan School operations management + CSAIL optimization

I created an AI-native supply chain platform integrating Google's Temporal Fusion Transformer with quantum-inspired QUBO optimization. The system achieves 94.8% forecast coverage with conformal prediction guarantees, uses offline reinforcement learning for policy improvement, and includes real-time drift detection. Enterprise features include S3-backed persistence, Monte Carlo financial simulation, and an LLM-powered analytics copilot.

Key Outcomes:
• 94.8% coverage guarantee on demand forecasting
• Quantum QUBO solver for NP-hard inventory optimization
• Full cloud-native deployment with S3 persistence

Accessibility Computing | Developer

2023 • 10 hrs/week • 200 total hours
Aligns with MIT's: CSAIL Accessibility Group human-centered computing

I built an application enabling computer operation for users without hands—tracking nose movement for cursor control and integrating speech recognition for click commands. The system processes webcam input at 30fps with <100ms latency, making standard computing accessible to users with upper-limb disabilities.

Key Outcomes:
• <100ms latency nose-tracking cursor control
• Speech recognition integration for 12 command types
• Deployed to 3 disability advocacy organizations

Data Science Challenge | Team Leader, NUS Program

December 2023 • 40 hrs/week • 80 total hours
Aligns with MIT's: Collaborative problem-solving culture

I led a 5-person team through a customer churn prediction challenge, pioneering AWS adoption within our cohort. I implemented Random Forest, SVM, and Neural Network models, achieving 99.8% accuracy through systematic hyperparameter optimization and ensemble methods.

Key Outcomes:
• 99.8% prediction accuracy (top 3 in cohort)
• First team to deploy AWS infrastructure in program
• Led 5-person team through 2-week intensive

Hackathon Champion | Indus International School

2022 • 40 hrs/week • 40 total hours
Aligns with MIT's: Hands-on innovation under constraints

In 24 sleepless hours, I built a gesture-controlled computer interface using webcam-based hand tracking. The system enabled cursor control through finger movements, winning first place against 30+ teams for technical innovation and practical accessibility applications.

Key Outcomes:
• 1st place among 30+ competing teams
• Real-time gesture recognition at 25fps
• Built complete prototype in 24 hours

ARIMA Inventory Forecasting Research | Researcher, IEEE

2024 • 20 hrs/week • 520 total hours
Aligns with MIT's: ORC statistical forecasting foundations

I conducted systematic research on ARIMA models for inventory forecasting, analyzing historical data patterns and developing forecasting frameworks. This foundational work informed my later quantum-classical integration research.

Key Outcomes:
• Analyzed 5 years of inventory data across 200+ SKUs
• Developed ARIMA model selection framework
• Published findings through IEEE

IBM Full Stack Development | Certified Developer

2024 • 20 hrs/week • 160 total hours
Aligns with MIT's: End-to-end systems building

Completed comprehensive certification building full-stack applications with React, Node.js, Express, MongoDB, and IBM Cloud deployment. Implemented CI/CD pipelines and completed 4 production-grade projects.

Key Outcomes:
• 4 full-stack applications deployed to production
• Proficiency across 8 frameworks/technologies
• CI/CD pipeline implementation

DSA Engineering Internship | Software Engineering Intern

June–August 2024 • 40 hrs/week • 320 total hours
Aligns with MIT's: Algorithmic problem-solving foundations

Applied data structures and algorithms to production challenges, solving 150+ problems and implementing optimized solutions for real-world engineering tasks.

Key Outcomes:
• Solved 150+ algorithmic challenges
• Implemented 12 production optimizations
• Reduced critical path runtime by 40%

Community Service Leadership | Member, Rotary Club

September 2022–September 2024 • 4 hrs/week • 208 total hours
Aligns with MIT's: Service-oriented leadership

Organized donation drives collecting 2,000+ items for underprivileged families, coordinated 8 community outreach events, and led awareness campaigns reaching 500+ community members.

Key Outcomes:
• Collected 2,000+ items through donation drives
• Coordinated 8 community outreach events
• Reached 500+ community members

Youth Mentorship | Mentor

2023–2024 • 2 hrs/week • 80 total hours
Aligns with MIT's: Knowledge transfer and community building

Mentored 12 younger students in academic and personal development, with 10 reporting improved academic performance and confidence.

Key Outcomes:
• Mentored 12 students over 40 weeks
• 83% reported improved academic performance

Sustainable Agriculture Education | Participant, Green Tokri

2023 • 2 hrs/week • 40 total hours
Aligns with MIT's: Environmental sustainability

Learned organic farming techniques emphasizing resource efficiency, implementing sustainable tree-growing practices that reduced water usage by 30%.

Key Outcomes:
• Implemented 30% water reduction techniques
• Grew 15 trees using sustainable methods

HONORS & RECOGNITION

• **Silver Medalist, 5th International STEM Olympiad** — Global recognition (2025)
• **National-Level Chess Player** — Top 200 in India, Under-19 category
• **2-Time State Champion & 2-Time District Champion** — Chess
• **IEEE Research Publications** — Quantum supply chain optimization, ARIMA forecasting (2025)

TECHNICAL SKILLS

ML/AI: TensorFlow, PyTorch, XGBoost, Reinforcement Learning, Quantum Computing (QUBO/QAOA)
Full Stack: React, Node.js, FastAPI, MongoDB, PostgreSQL, Docker, AWS
Research: Statistical modeling, Tensor decomposition, Conformal prediction
Languages: English (Native), Hindi (Native)

WHY MIT

My quantum-classical supply chain research directly extends Prof. Dimitris Bertsimas's prescriptive analytics work at the Operations Research Center—I want to formalize quantum decision representations within his optimization frameworks. MIT's unique integration of CSAIL's systems-building culture with Sloan's operations management expertise matches my approach of bridging theoretical rigor with deployed solutions. The MIT-IBM Watson AI Lab's focus on enterprise AI deployment aligns with my SAP experience translating research into production systems. Most importantly, MIT's "Mens et Manus" philosophy—using technical mastery for real-world impact—reflects my work building accessibility technology and education systems for Afghan girls. I want to pursue Course 15 (Management Science) with ORC affiliation, combining optimization theory with the hands-on building that defines MIT.

ADDITIONAL EXPERIENCE

Intern | NA

2024-06 - 2024-08 • 320 total hours

Demonstrated strong skills in Data Structures and Algorithms (DSA). Exhibited effective problem-solving abilities. Assigned various tasks and successfully solved them. Applied theoretical knowledge to practical challenges. Enhanced technical skills through hands-on experience. Reinforced capacity to solve complex problems efficiently.

Student | IBM

NA - NA • 160 total hours

Completed IBM Certified Full Stack Development course with hands-on experience in both front-end and back-end technologies. Developed dynamic, responsive web applications using modern tools and frameworks. Front-End Development: Proficient in HTML, CSS, JavaScript, React, and Bootstrap. Built user-friendly, visually appealing interfaces. Back-End Development: Experienced with Node.js, Express.js, Python, and RESTful APIs. Implemented efficient server-side logic and data management. Databases: Worked with MongoDB and SQL for data storage, retrieval, and management. Version Control: Proficient in Git and GitHub for version control and collaboration. Deployment: Familiar with cloud services such as IBM Cloud and implemented CI/CD pipelines for deployment. Completed multiple real-world projects focusing on problem-solving and software development best practices. Gained experience in full-stack web application design, development, and maintenance. Confident in contributing to both client-side and server-side development.

Researcher | IEEE

Unknown - Unknown • 520 total hours

Conducted research on using ARIMA (Autoregressive Integrated Moving Average) models for inventory forecasting. Analyzed historical data, developed forecasting models, and evaluated the accuracy and effectiveness of the ARIMA approach.

Researcher |

- Present • 0 total hours

Proposes a unified framework combining ARIMA demand forecasting, quantum decision theory, and reinforcement learning for omnichannel retail.

Models customer intent as quantum superpositions to capture contextual and cross-channel dependencies absent in classical models.

Embeds forecast uncertainty directly into decision states and reward structures.

Demonstrates improved profitability (12–18%), reduced stock-outs, and higher service levels under demand volatility and promotions.

Researcher |

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